Ten Ways to Use Agentic AI in Academic Research

Short, standalone blurbs describing concrete uses for AI coding agents (Claude Code, Cursor, and similar) in academic work: the kinds of tasks covered in the workshop.

1. Literature triage and synthesis

Upload a stack of PDFs and ask an agent to summarize each, extract the core claim and method, and compare them against your own thesis. Agents are especially useful for the "is this paper actually relevant to me?" triage that eats hours of reading time, and for building a first-pass map of an unfamiliar literature before you dive in yourself. The point is not to replace close reading of the papers that matter, but to cut the pile of "maybe relevant" papers down to the ones that truly are.

2. Writing empirical code

From data cleaning to model specifications, agents can write, run, and debug pandas, R, and Stata code. The productivity gain is largest for tedious-but-mechanical work: reshaping panels, merging messy datasets, producing descriptive statistics, and iterating on figure aesthetics until they are submission-ready. A task that used to mean an afternoon of StackOverflow becomes a five-minute conversation, which frees you to spend time on the parts of the analysis that actually require judgment.

3. Replication checks

Point an agent at an author's replication package and ask whether the tables and figures actually reproduce. This is invaluable both as a referee — catching silent errors before publication — and as a student learning a new method, because you get to see exactly how a published result was built from raw data to final table. It's also a humbling exercise on your own work: run it on your own replication package before you submit.

4. Specification and robustness checks

Before you run an analysis — quantitative or qualitative — ask the agent to audit your setup: how you handled missing data, how your sample is defined, how your variables or codes are constructed, whether your unit of analysis matches the claim you want to make. Once you have results, have it stress-test them against alternative samples, recodings, and modeling choices. This catches silent errors far more reliably than self-review and produces findings you can defend with confidence at a seminar or in a referee report.

5. Drafting and editing prose

Abstracts, introductions, response-to-referee letters, grant narratives, cover letters, job-talk intros. Agents are strong at tightening flabby prose without changing your argument, reorganizing sections for flow, and flipping register between technical and lay audiences, which is useful for the same paper being pitched to a field journal, a general audience, and a funder. Used well, an agent is a patient editor who will read your eighth draft at 11pm without complaint.

6. Red-teaming your own paper

Ask the agent to write a skeptical referee report on your draft before you submit it. Have it stress-test your identification strategy, hunt for alternative explanations, and flag the claims your evidence doesn't actually support. This surfaces weaknesses while you still have time to fix them, rather than learning about them six months later from Reviewer 2. It works equally well on a grant proposal, a job-market paper, or a chapter draft.

7. Teaching materials

Lecture slides, problem sets, exam questions, TA guides, syllabi. Agents can adapt the same content across difficulty levels (undergrad to PhD), generate fresh variants of problems to discourage copying from prior-year solution sets, and draft worked solutions you can edit. This is especially valuable when you are preparing a new course where you are building everything from scratch, and the marginal slide or problem is the difference between a polished class and a rough one.

8. LaTeX and figure tooling

Regression tables formatted to journal specs, Beamer slide layouts, BibTeX cleanup, tikz diagrams, and debugging the obscure compile errors that appear the morning a submission is due. LaTeX has a steep learning curve and a long tail of weird failure modes; agents are fluent in both and save hours of StackOverflow. The same holds for matplotlib, ggplot, and the unreasonable amount of time academics spend wrestling legends and axis labels.

9. Research decision logging and replication documentation

As the project evolves, an agent can automatically maintain a README, codebook, and decision log capturing every variable construction, sample restriction, and specification choice *when it's made* — not reconstructed from memory months later during revision. This is the single highest-leverage habit for minimizing the pain of R&Rs and making your work genuinely replicable. Your future self, and anyone who inherits the project, will thank you.

10. Learning new methods or unfamiliar literatures

When you need to come up to speed on something new — a statistical technique, an archival approach, a theoretical framework, a body of scholarship outside your field — an agent can explain it, walk through a worked example, point you to the canonical references, and critique your planned application. It is the academic equivalent of having a patient senior colleague on call, and it is especially valuable for solo researchers, people between institutions, or anyone whose local peer group doesn't happen to cover the method they need this week.